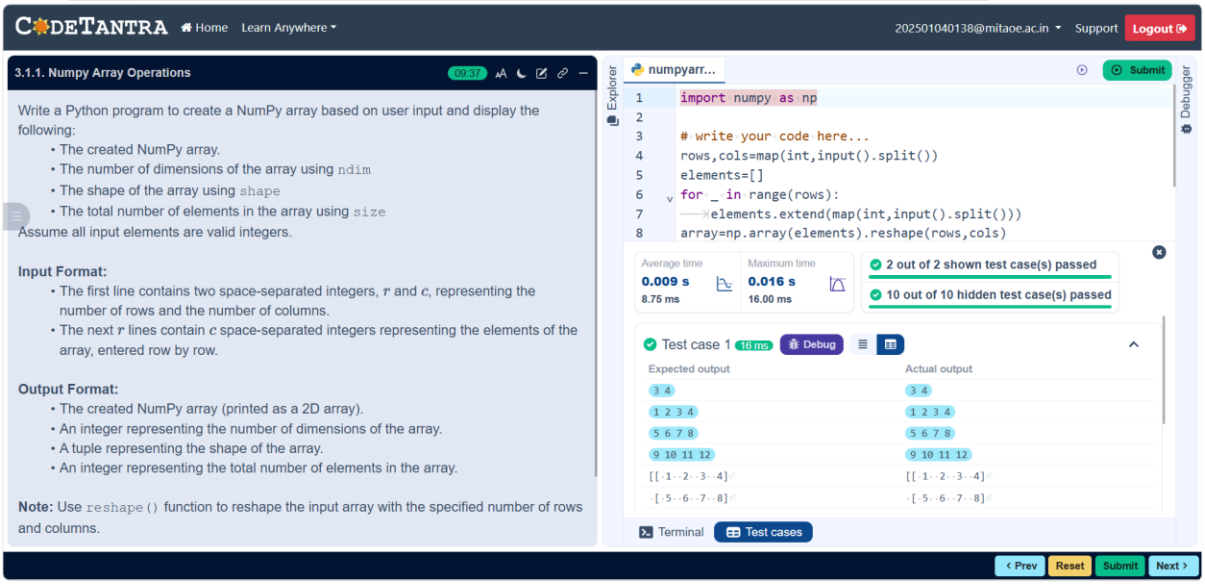
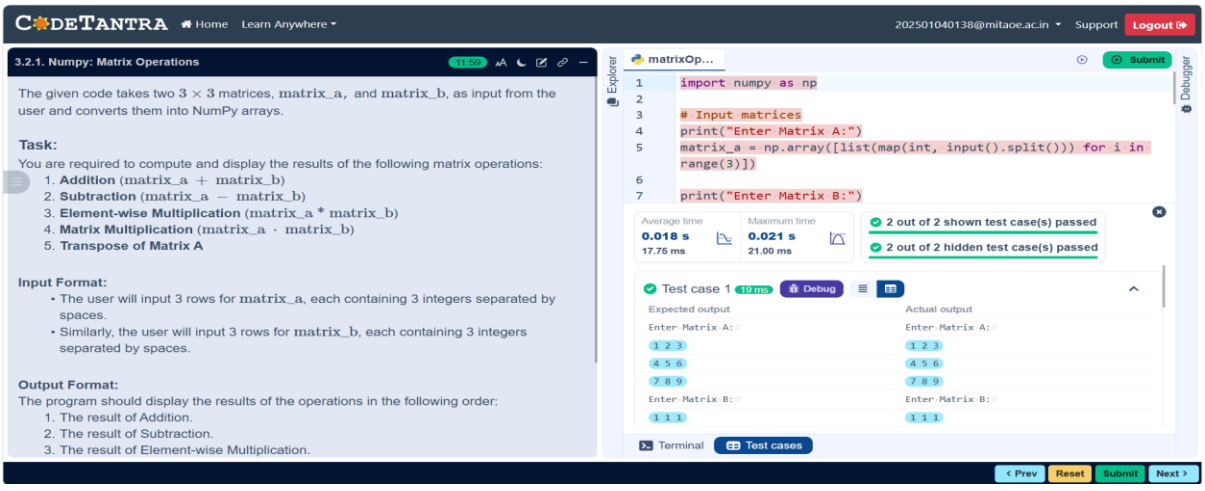
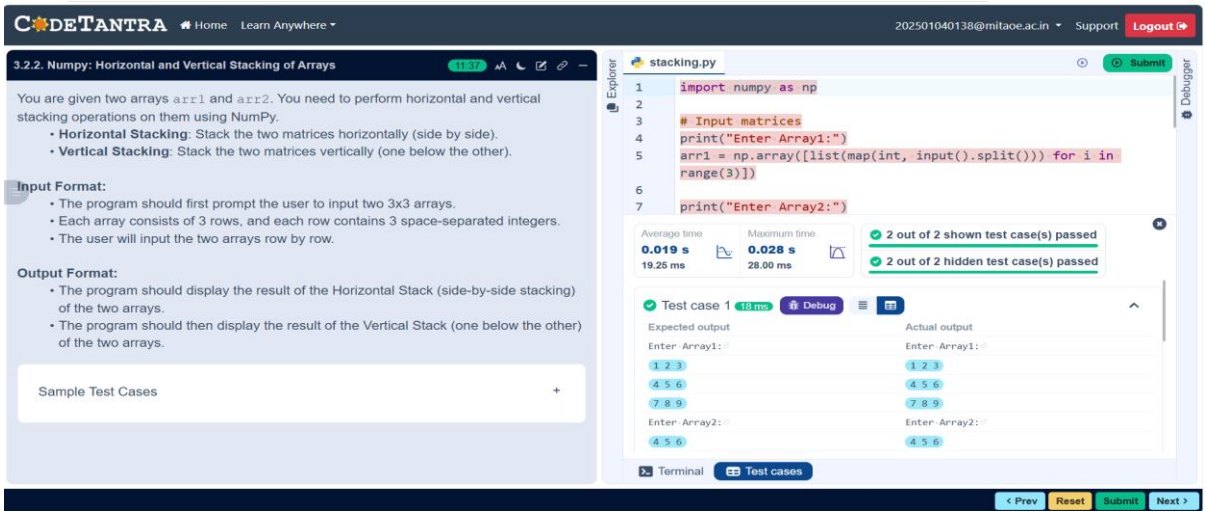
Practical 3

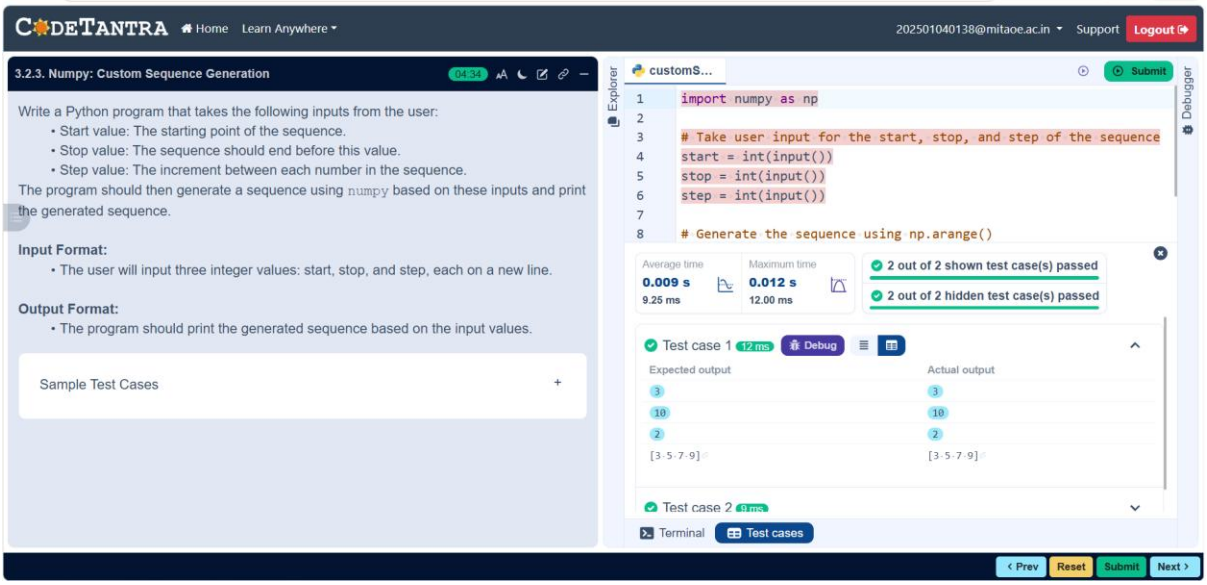
Practice Lab Assignment

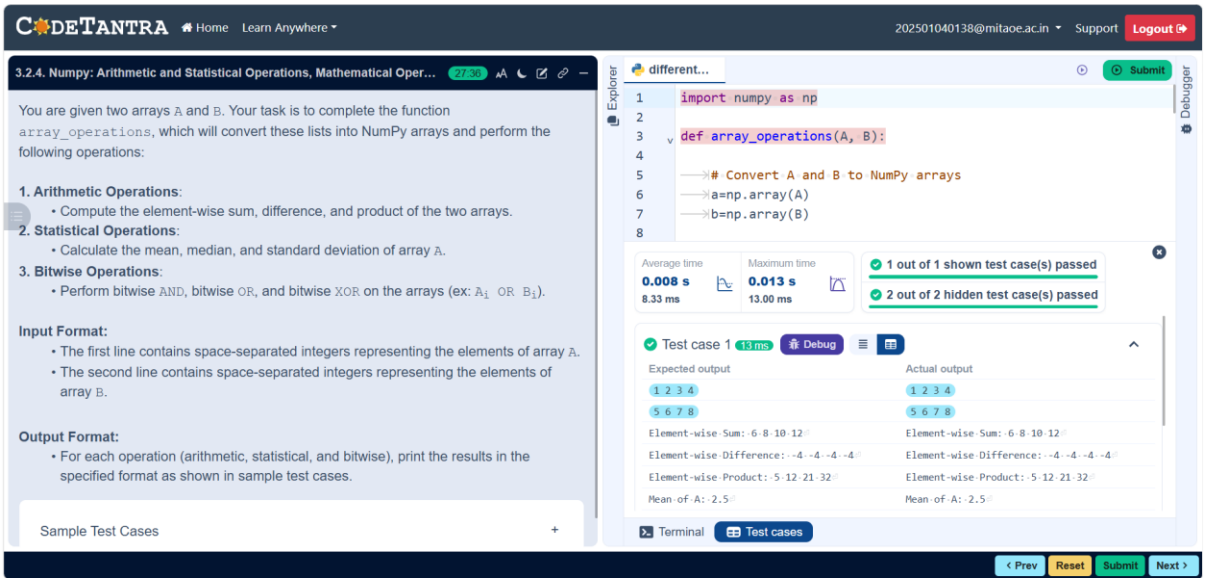


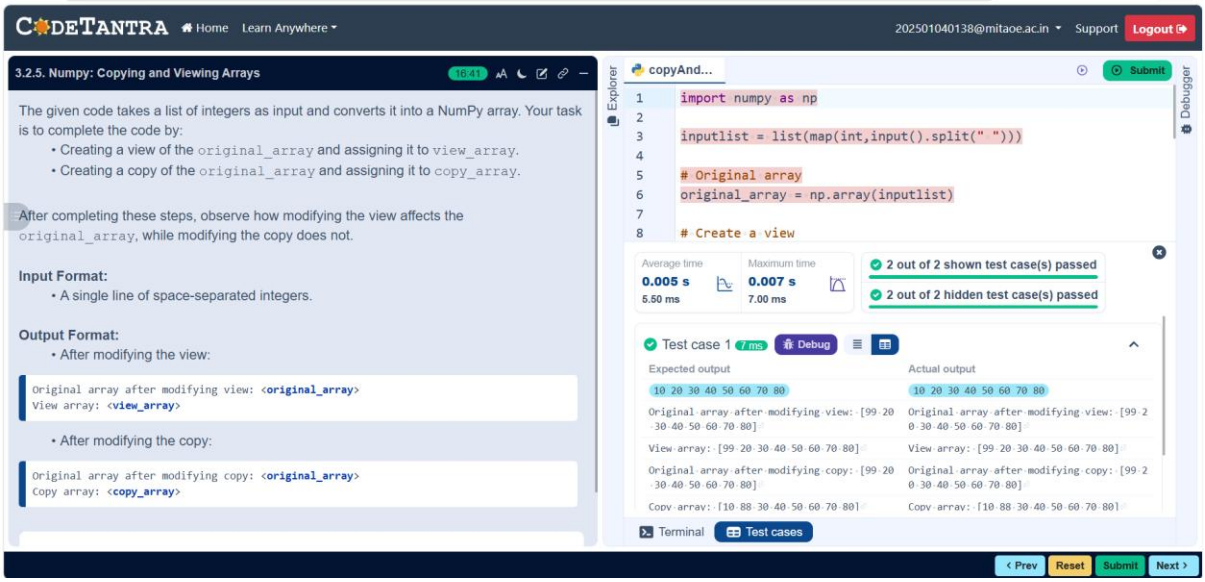
Lab Assignment

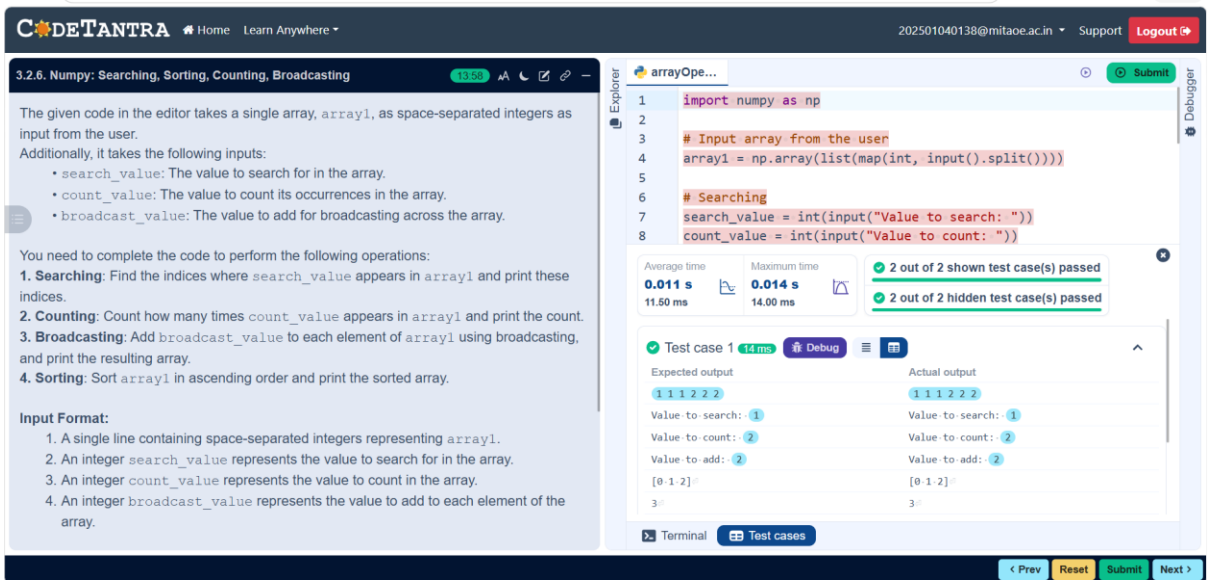


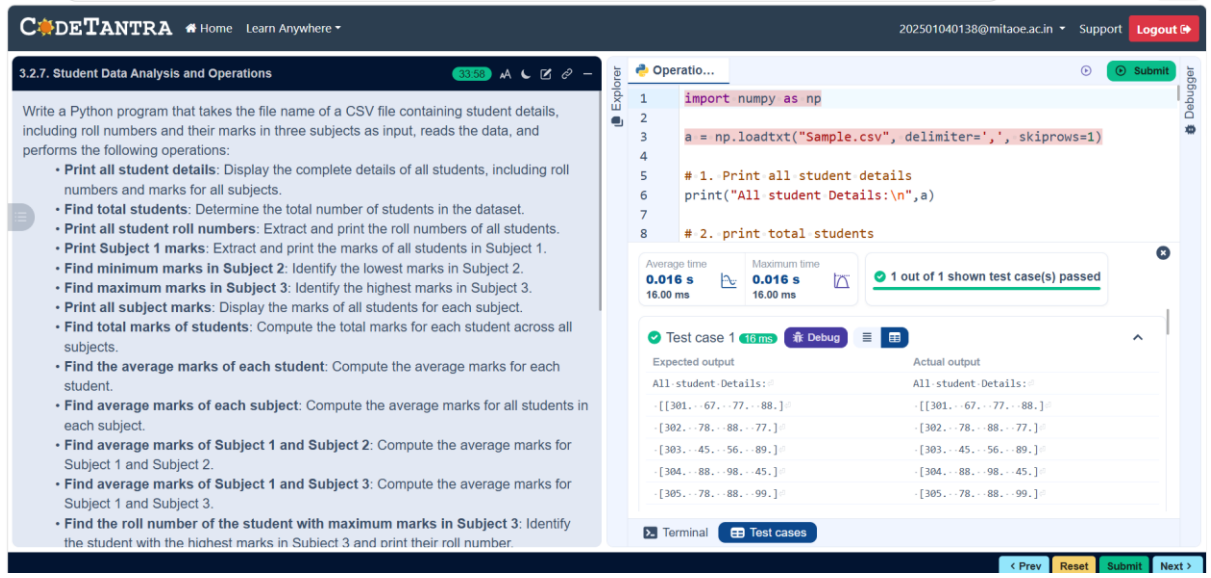



CODETANTRA
Home
Learn Anywhere
202501040138@mitaoe.ac.in
Support
Logout
3.2.3. Numpy: Custom Sequence Generation
04:34
Write a Python program that takes the following inputs from the user:
• Start value: The starting point of the sequence.
• Stop value: The sequence should end before this value.
• Step value: The increment between each number in the sequence.
The program should then generate a sequence using numpy based on these inputs and print the generated sequence.
Input Format:
• The user will input three integer values: start, stop, and step, each on a new line.
Output Format:
• The program should print the generated sequence based on the input values.
Sample Test Cases
customS...
Submit
Explorer
Debugger
import numpy as np
# Take user input for the start, stop, and step of the sequence
start = int(input())
stop = int(input())
step = int(input())
# Generate the sequence using np.arange()
Average time
0.009 s
9.25 ms
Maximum time
0.012 s
12.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
12 ms
Debug
Expected output
Actual output
3
10
2
[3 5 7 9]
Test case 2
Terminal
Test cases
Prev
Reset
Submit
Next


CODETANTRA
Home
Learn Anywhere
202501040138@mitaoe.ac.in
Support
Logout
3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Oper...
27:30
You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:
1. Arithmetic Operations:
• Compute the element-wise sum, difference, and product of the two arrays.
2. Statistical Operations:
• Calculate the mean, median, and standard deviation of array A.
3. Bitwise Operations:
• Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).
Input Format:
• The first line contains space-separated integers representing the elements of array A.
• The second line contains space-separated integers representing the elements of array B.
Output Format:
• For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.
Sample Test Cases
different...
Submit
Explorer
Debugger
import numpy as np
def array_operations(A, B):
# Convert A and B to NumPy arrays
a=np.array(A)
b=np.array(B)
Average time
0.008 s
8.33 ms
Maximum time
0.013 s
13.00 ms
1 out of 1 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
13 ms
Debug
Expected output
Actual output
1 2 3 4
5 6 7 8
Element-wise Sum: 6 8 10 12
Element-wise Difference: -4 -4 -4 -4
Element-wise Product: 5 12 21 32
Mean of A: 2.5
Terminal
Test cases
Prev
Reset
Submit
Next


CODETANTRA
Home
Learn Anywhere
202501040138@mitaoe.ac.in
Support
Logout
3.2.5. Numpy: Copying and Viewing Arrays
16:41
The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:
• Creating a view of the original_array and assigning it to view_array.
• Creating a copy of the original_array and assigning it to copy_array.
After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.
Input Format:
• A single line of space-separated integers.
Output Format:
• After modifying the view:
Original array after modifying view: <original_array>
View array: <view_array>
• After modifying the copy:
Original array after modifying copy: <original_array>
Copy array: <copy_array>
copyAnd...
Submit
Explorer
Debugger
import numpy as np
inputlist = list(map(int,input().split(" ")))
# Original array
original_array = np.array(inputlist)
# Create a view
Average time
0.005 s
5.50 ms
Maximum time
0.007 s
7.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
7 ms
Debug
Expected output
Actual output
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
Terminal
Test cases
Prev
Reset
Submit
Next


CODETANTRA
Home
Learn Anywhere
202501040138@mitaoe.ac.in
Support
Logout
3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting
13:58
The given code in the editor takes a single array, array1, as space-separated integers as input from the user.
Additionally, it takes the following inputs:
• search_value: The value to search for in the array.
• count_value: The value to count its occurrences in the array.
• broadcast_value: The value to add for broadcasting across the array.
You need to complete the code to perform the following operations:
1. Searching: Find the indices where search_value appears in array1 and print these indices.
2. Counting: Count how many times count_value appears in array1 and print the count.
3. Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. Sorting: Sort array1 in ascending order and print the sorted array.
Input Format:
1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.
Explorer
arrayOpe...
Submit
Debugger
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split())))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
Average time
0.011 s
11.50 ms
Maximum time
0.014 s
14.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
14 ms
Debug
Expected output
1 1 1 2 2 2
Value to search: 1
Value to count: 2
Value to add: 2
[0 1 2]
3
Actual output
1 1 1 2 2 2
Value to search: 1
Value to count: 2
Value to add: 2
[0 1 2]
3
Terminal
Test cases
Prev
Reset
Submit
Next


CODETANTRA
Home
Learn Anywhere
202501040138@mitaoe.ac.in
Support
Logout
3.2.7. Student Data Analysis and Operations
33:58
Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:
• Print all student details: Display the complete details of all students, including roll numbers and marks for all subjects.
• Find total students: Determine the total number of students in the dataset.
• Print all student roll numbers: Extract and print the roll numbers of all students.
• Print Subject 1 marks: Extract and print the marks of all students in Subject 1.
• Find minimum marks in Subject 2: Identify the lowest marks in Subject 2.
• Find maximum marks in Subject 3: Identify the highest marks in Subject 3.
• Print all subject marks: Display the marks of all students for each subject.
• Find total marks of students: Compute the total marks for each student across all subjects.
• Find the average marks of each student: Compute the average marks for each student.
• Find average marks of each subject: Compute the average marks for all students in each subject.
• Find average marks of Subject 1 and Subject 2: Compute the average marks for Subject 1 and Subject 2.
• Find average marks of Subject 1 and Subject 3: Compute the average marks for Subject 1 and Subject 3.
• Find the roll number of the student with maximum marks in Subject 3: Identify the student with the highest marks in Subject 3 and print their roll number.
Explorer
Operatio...
Submit
Debugger
1 import numpy as np
2
3 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
4
5 # 1. Print all student details
6 print("All student Details:\n",a)
7
8 # 2. print total students
Average time
0.016 s
16.00 ms
Maximum time
0.016 s
16.00 ms
1 out of 1 shown test case(s) passed
Test case 1
16 ms
Debug
Expected output
All student Details:
[[301. 67. 77. 88.]
[302. 78. 88. 77.]
[303. 45. 56. 89.]
[304. 88. 98. 45.]
[305. 78. 88. 99.]]
Actual output
All student Details:
[[301. 67. 77. 88.]
[302. 78. 88. 77.]
[303. 45. 56. 89.]
[304. 88. 98. 45.]
[305. 78. 88. 99.]]
Terminal
Test cases
Prev
Reset
Submit
Next